

Nigam Niraula

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Professional Experience

2025/11 – Present	SecurityPal AI <i>Research and Development Engineer (Full Time)</i> - Designed and implemented a diffusion-based text generation model that probabilistically reasons over contextual embeddings to answer complex, context-aware questions. - Built a fully autonomous, end to end report generation pipeline that transforms raw, unstructured data into coherent, publication ready narratives and charts with zero human intervention.
2025/05 – 2025/11 Kathmandu, Nepal	SecurityPal AI <i>Associate Machine Learning Engineer (Full Time)</i> - Built agentic AI systems using Google ADK, focusing on modular agent design and goal-driven behavior. - Integrated A2A (Agent-to-Agent) communication to enable autonomous coordination and task execution among agents. - Implemented a graph-based natural language representation framework to enhance retrieval precision and context awareness in Retrieval-Augmented Generation (RAG) systems.
2025/09 – Present Lalitpur, Nepal	The Westminster College <i>Visiting Faculty (Part Time)</i> - Taught Python programming fundamentals to first-year Computer Science students, covering core concepts such as control flow, data structures, and problem solving. - Delivered Software Development instruction to second-year students, focusing on modern front-end and back-end practices.
2024/08 – 2025/05 Hyderabad, India	AI Lab IITH  <i>Research Intern (Part Time)</i> Worked on Diffusion Model, Conditional Diffusion Models and Indian Styled Content Generation using Diffusion Models
2023/11 – 2024/08 Dhulikhel, Nepal	Kathmandu University Active Learning Lab <i>Research Intern (Part Time)</i> Worked on image detection using CNN and image generation for a given tag using GAN

Education

2020/03 – 2025/05 Dhulikhel, Nepal	Bachelors in Computer Engineering <i>Kathmandu University School of Engineering</i> CGPA 3.96/4 Academic Exchange Student (Final Year): Indian Institute of Technology, Hyderabad (IIT-H) Hyderabad, India 2024/08 – 2025/05 - Completed final year B.Tech  coursework and projects. Exchange CGPA: 8.83/10
2018 – 2020 Bhaktapur, Nepal	+2 Science <i>SOS Hermann Gmeiner School Sanothimi</i> CGPA: 3.53

Publication

MIDAS: Adaptive Proxy Middleware for Mitigating Metadata Hotspots in HPC I/O at Scale 
S. Ghimire, N. Niraula, N. Bhurtel, P. Timalisina, B. Neupane, J. Bhattarai, and S. Jha, "MIDAS: Adaptive Proxy Middleware for Mitigating Metadata Hotspots in HPC I/O at Scale," in Proceedings of the 16th IEEE International Conference on Cloud Computing Technology and Science (CloudCom 2025), Shenzhen, China, 2025. [Preprint available: [arXiv:2511.18124](https://arxiv.org/abs/2511.18124) 

Hybrid Approach for Enhancing Traffic Signal Control with Computer Vision and Reinforcement Learning [↗](#)

B. Neupane, N. Niraula, N. Bhurtel, J. K. Magar, and P. R. Dawadi, "Hybrid Approach for Enhancing Traffic Signal Control with Computer Vision and Reinforcement Learning," in *14th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Delhi, India, 2023.

Skills

Technical

TensorFlow, PyTorch, Python, Scikit, Keras, Numpy, Pandas, SQL, Javascript, React, NodeJs, C++, GIT, Bash

Research

Deep Learning, Machine Learning, Computer Vision

Projects

Autoffic [↗](#)

A software for automating the traffic flow of an intersection by controlling the traffic lights via the input through **Computer Vision** and output through **RL**.

Prabigya [↗](#)

VS code extension to suggest variable names based on the comment by programmers for enhancing readability of program by using **LLM (Bard)**.

Farm App [↗](#)

Farm App is a inventory management app made using **React Native** for chicken farms to manage eggs, chicken and chicken's feed, water, light, etc.

Ayureye [↗](#)

Ayureye is a CAD (Computer aided Detection and Diagnosis) system that detects COPD (Common Obstructive Pulmonary Disease) implemented using **ReactJs, Django** and **YOLO** architecture.

Prophesy [↗](#)

A battery monitoring system powered by **RUST** and **ML** for motoring battery status and predicting the battery status using neural network and regression algorithms

SwiftCart [↗](#)

An e-commerce application that provides a platform for vendors to list their products, customers to browse and purchase items, and facilitate reviews for products utilizing **MERN**.

Certificates

Microdegree™ in Artificial Intelligence [↗](#)

Fusemachines AI Fellowship 2024

Programming Fundamentals

Duke University, Coursera

Supervised Machine Learning: Regression and Classification

Dr. Andrew NG, Coursera

Effective Public Speaking Training(EPS)

JCI Kavre

Leadership and Activities

- Recipient of AI Fellowship by Fusemachine
- Selected in top 50 participants of Kathmandu Idea and Innovation(KMII)-2080
- Participated as panelist in Scholar Talk-2023 held by Kathmandu University Civil Engineering Club(KUCEC)